

Sandhya Lotla

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Summary

Data Engineer & Analyst with over 2.5 years of experience building scalable ETL pipelines and delivering actionable insights. Proficient in SQL, Python, and cloud platforms, with a strong record in designing self-service BI dashboards and reports. Adept at collaborating with cross-functional teams to ensure data integrity and drive improvements in key performance indicators.

Education

University of New Haven

Master's, Data Science

Jan 2024 – Dec 2025

Guru Nanak Institute of Technology

Bachelor's, Computer Science & Engineering

Jul 2017 - Sep 2021

Skills

- **Programming Languages:** Python, SQL, R
- **Data Analysis & Statistics:** Pandas, NumPy, Statistical Analysis, A/B Testing, SAS, SPSS, Predictive Modeling
- **Data Visualization & BI Tools:** Tableau, Power BI, Matplotlib, Seaborn, Looker, QlikView
- **Machine Learning:** Scikit-learn, XGBoost, Logistic Regression, Random Forest, Model Evaluation
- **ETL & Data Engineering:** AWS Glue, PySpark, Apache Airflow, Azure Data Factory, Batch/Stream Processing
- **Cloud Platforms:** AWS (S3, Lambda, Redshift), Snowflake, Azure (Synapse, Data Factory), Databricks
- **Databases & Warehousing:** MySQL, Google BigQuery, Amazon Redshift, PostgreSQL, Oracle
- **Tools & Collaboration:** Git, Jira, Jenkins, Excel (Power Pivot, VBA, Advanced), Docker

Experience

Tata Consultancy Services (TCS)

Data Analyst

Aug 2021 - Jan 2024

Bangalore, India

- Collaborated with engineering and business teams to design MySQL database schemas for customer review data storage, optimizing OLTP performance with indexing and partitioning, and reducing query latency by 18%.
- Built robust ETL pipelines using SQL to aggregate and transform raw data from multiple sources, enabling seamless integration into AWS S3 data lakes and ensuring data quality through stringent validation checks.
- Engineered batch processing workflows using Python and Hadoop to ingest and clean both structured and unstructured data from MySQL and MongoDB, addressing schema mismatches and ensuring data consistency for downstream analytics.
- Supported BI initiatives by analyzing marketing campaign metrics (e.g., CTR, conversion rates) with R, developing automated weekly reports that reduced manual effort by 10 hours per month and provided actionable insights.
- Developed interactive Tableau dashboards to visualize customer sentiment trends, empowering stakeholders to improve key performance indicators and achieving a 15% improvement in campaign ROI.
- Implemented feature engineering pipelines using Python to prepare data for machine learning models, including normalization and PCA-driven dimensionality reduction, enhancing model training efficiency by 25%.
- Contributed to machine learning model development with scikit-learn, training and validating Logistic Regression and Random Forest classifiers to achieve 88% accuracy in customer churn prediction.
- Orchestrated data workflows with Apache Airflow to automate ETL jobs, ensuring SLA-compliant daily data refreshes and reliable delivery of insights for BI reporting.
- Maintained data security through robust access control measures, including field-level and role-based security protocols.
- Participated in Agile project cycles, collaborating in daily stand-ups and sprint planning to ensure timely delivery of data analytics and BI improvements.

Projects

Optimizing Tweet Sentiment Analysis with EC2 and Airflow

- Designed and implemented a scalable ETL pipeline using Apache Airflow on AWS EC2 for sentiment classification.
- Automated data ingestion using Kaggle API, transforming raw tweets into structured sentiment labels.
- Employed TF-IDF and count vectorization for feature engineering and optimized feature selection with PCA.
- Trained models (Logistic Regression, Random Forest, Naive Bayes, SVM) and evaluated using precision, recall, F1-score.
- Visualized sentiment distribution trends using Power BI and shared results via real-time dashboards.
- Orchestrated daily model retraining using Airflow DAGs and EC2 for scalability, aligned with real-world streaming needs.
- Applied best practices in code versioning using Git and Dockerized pipelines for deployment readiness.

Gender and Age Detection from Facial Images

- Built classification pipeline using OpenCV for face detection and Gradio for live UI interaction.
- Applied transfer learning with pre-trained CNNs to detect 8 age categories and binary gender classification.
- Handled challenges in facial variability (lighting, expression, obstruction) via advanced image augmentation.
- Implemented logging and result tracking for model evaluation using TensorBoard.
- Designed modular codebase in Python for reproducibility and deployment.
- Explored integration potential for HR biometric applications and access control use cases.

Academic Project

VINIMI (Vision and Natural Language Intelligence for Monitoring Industrial Safety)

- Developed an AI-driven system integrating computer vision and natural language processing (NLP) for intelligent safety monitoring and reporting.
- Implemented helmet detection using YOLOv8 and face recognition with DeepFace + OpenCV, achieving >92% accuracy in PPE compliance detection.
- Designed a synthetic dataset pipeline using Stable Diffusion, ensuring scalable and diverse training data without manual collection.
- Automated SMS alerts with Twilio for non-compliance (no_helmet), integrating metadata lookup for worker ID and contact mapping.
- Built a user-friendly Streamlit/Gradio UI to allow image upload and natural queries (e.g., "Is the worker wearing a helmet?") powered by LLaVA/BLIP-2 vision-language models.
- Evaluated model performance using machine learning algorithms and statistical analysis techniques, enhancing accuracy by 12%.
- Environment: Python, YOLOv8, DeepFace, OpenCV, Stable Diffusion, Twilio, Streamlit, Gradio, AWS Certification

Certifications

- Power BI Certification
- AWS Certified Solutions Architect - Associate
- Google Data Analytics Certificate
- Completed Deep Learning Specialization (Coursera)